Ethical Compliance Declaration

AI-Powered Chatbot for University Student Support — UniAssist

HCLTech Career Shaper Programme — Digital-Data-AI / AIML Domain

Declaration

I hereby declare that the UniAssist project — an AI-Powered Chatbot for University Student Support — has been developed in full compliance with ethical standards, data privacy regulations, and responsible AI principles as outlined by the HCLTech Career Shaper Programme and applicable institutional guidelines.

1. Data Privacy & Student PII Protection

Principle | Implementation

No PII Storage Beyond Session | Student personally identifiable information (PII) including phone numbers, email addresses, and identification numbers is automatically masked using regex-based scrubbing before any data is persisted to the database. Session memory is cleared after a 30-minute time-to-live (TTL) window.

PII Masking Patterns | Indian mobile numbers (`+91`/`0` prefix, 10 digits) → `[PHONE_MASKED]`; Email addresses → `[EMAIL_MASKED]`; Aadhaar numbers (12-digit format) → `[ID_MASKED]`

No External Data Sharing | Student queries and responses are stored only in the local SQLite database for analytics purposes. No student data is transmitted to third-party services beyond the LLM API for generating responses.

Minimal Data Collection | Only session-level metadata (anonymized queries, response quality ratings, escalation tickets) is retained. No student names, enrollment numbers, or demographic data is stored in chat logs.

2. Opt-Out Mechanism

- Students are **not required** to provide any personal information to use the chatbot.
- The chatbot operates on an **anonymous session basis** — no login or authentication is required for student users.
- Students may **clear their session** at any time using the "New Session" or "Clear Chat" controls, which immediately purges all conversation history from memory.
- The **"Talk to Human"** escalation feature is entirely voluntary — students choose whether to provide their name, student ID, and email when submitting a support ticket.

3. Human Escalation Transparency

• The chatbot **clearly indicates** when a query cannot be resolved automatically and provides explicit options for human escalation.

• Escalation tickets are generated with unique identifiers (`TICK-YYYYMMDD-XXXX`) and visible confirmation messages.

• Students receive clear communication that their query will be reviewed by a human support agent.

• The escalation button ("Talk to Human") is **prominently visible** on the chat interface at all times, ensuring students always have the option to reach a human agent.

• Automated responses clearly state when the chatbot is unable to provide a definitive answer and directs students to official university contacts.

4. Responsible AI & Safety Guardrails

Guardrail | Purpose

Prompt Injection Defense | Prevents jailbreak attempts, system prompt extraction, and API key probing through regex pattern matching.

Profanity & Abuse Filtering | Maintains a respectful and professional interaction environment by detecting and deflecting abusive language.

Out-of-Scope Detection | Politely redirects queries unrelated to university topics (e.g., stock trading, entertainment) to prevent misuse.

Hallucination Prevention | All responses are grounded in official university knowledge base documents with cited sources. Queries with low similarity scores trigger fallback responses with official contact information.

5. Bias Mitigation

• The chatbot treats all student queries equally regardless of the language style, phrasing, or topic complexity.

• The knowledge base covers a comprehensive range of university topics (12 policy documents) to ensure balanced information access across all student demographics.

• Sentiment analysis is applied uniformly to understand student satisfaction without discriminating based on query characteristics.

• Regular evaluation through the 30-question benchmark harness ensures consistent performance across all intent categories.

6. Transparency & Accountability

• The chatbot identifies itself as an **AI assistant** and does not impersonate a human agent.

• Source citations are provided with every response, allowing students to verify information independently.

• All chat interactions are logged (with PII masking) for quality assurance and audit purposes.

• The administrative dashboard provides transparency into chatbot performance metrics, satisfaction scores, and unresolved tickets.

7. Dataset Ethics

- The training/evaluation dataset (`dataset.csv`) contains **synthetic data only** — no real student queries or personal information is used.
- The dataset was generated using AI-based paraphrasing tools following the HCL-provided Dataset Generation Prompt.
- No copyrighted, proprietary, or sensitive educational records were used in dataset construction.

Student Name: _____

Date: _____

Signature: _____

This declaration is submitted as part of the HCLTech Career Shaper Programme — III Year Industry Project (Digital-Data-AI / AIML Domain).